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Experiment 2:

AIM: Study of Python Basic Libraries such as Statistics, Math, Numpy and Scipy

PROGRAM:

```
import numpy as np
arr=np.array([5,6,7])
print(arr)
arr=np.array([[1,2,3],[4,5,6]])
print(arr)
arr=np.array([[1,2,3,4,5],[6,7,8,9,10],[11,12,13,14,15],[16,17,18,19,20]])
print(arr)
sliced_arr=arr[:2,::2]
print(sliced_arr)
row_indices=[1,1,0,3]
column_indices=[3,2,1,0]
indexed_elements=arr[row_indices,column_indices]
print(indexed_elements)
a=np.array([[1,2],[3,4]])
b=np.array([[5,6],[7,8]])
print(a+1)
print(b-1)
print(a+b)
print(a.sum())
s="suhail"
print(s)
print(s[2])
print(s[-2])
print(s[1:3])
print(s[2:6:1])
print(s[2:6:2])
```

Output:

```
[5 6 7]
[[1 2 3]
 [4 5 6]]
[[ 1  2  3  4  5]
 [ 6  7  8  9 10]
 [11 12 13 14 15]
 [16 17 18 19 20]]
[[ 1  3  5]
 [ 6  8 10]]
```

```
[ 9  8  2 16]
[[2 3]
 [4 5]]
[[4 5]
 [6 7]]
[[ 6  8]
 [10 12]]
10
suhail
h
i
uh
hail
hi
```

Basic Operations on Array:

PROGRAM:

```
x=np.array([[10,20,30,40],[5,10,15,20],[2,4,6,8]])
a1=x.flatten()
print(a1)
a1[0]=99
print(a1)
print(x)
x=np.array([[10,20,30,40],[5,10,15,20],[2,4,6,8]])
a2=x.ravel()
a2[0]=99
print(x)
print(a2)
x=np.array([10,20,30,40])
print(x.dtype)
x=np.array([10,2.30,30,40])
print(x.dtype)
x=np.arange(5)
print(x)
y=np.arange(12).reshape(4,3)
```

print(y)

Output:

```
[10 20 30 40  5 10 15 20  2  4  6  8]
[99 20 30 40  5 10 15 20  2  4  6  8]
[[10 20 30 40]
 [ 5 10 15 20]
 [ 2  4  6  8]]
[[99 20 30 40]
 [ 5 10 15 20]
 [ 2  4  6  8]]
[99 20 30 40  5 10 15 20  2  4  6  8]
int64
float64
[0 1 2 3 4]
[[ 0  1  2]
 [ 3  4  5]
 [ 6  7  8]
 [ 9 10 11]]
```

PROGRAM:

```
a=np.array([[3,4],[6,7]])
b=np.array([[11,2],[1,2]])
print(a@b)
print(a*b)
a=np.array([3,4,5,6,7])
b=np.arange(5)
print(a)
print(b)
print(a-b)
print(b**3)
print(a<6)
```

Output:

```
[[37 14]
[73 26]]
[[33  8]
[ 6 14]]
[3 4 5 6 7]
[0 1 2 3 4]
[3 3 3 3 3]
[ 0  1  8 27 64]
[ True  True  True False False]
```

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Scipy Library:

PROGRAM:

```
from scipy import special as sp
a=sp.exp10(4)
b=sp.exp2(9)
c=sp.sindg(90)
print(a)
print(b)
print(c)
from scipy import linalg
x=np.array([[10,26],[5,10]])
print(x)
b=linalg.inv(x)
print(b)
```

Output:

```
10000.0
512.0
1.0
[[10 26]
[ 5 10]]
[[-0.33333333  0.86666667]
[ 0.16666667 -0.33333333]]
```

Differentiation using scipy library:

PROGRAM:

```
from scipy.differentiate import derivative
def my_function(x):
        return x**5+4*x+7
a=derivative(my_function,2)
print(a)
```

Output:

```
success: True
status: 0
df: 83.99999999999952
error: 1.9895196601282805e-13
nit: 2
nfev: 11
x: 2.0
```

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Eigen Values and Eigen Vectors using scipy library:

PROGRAM:

```
arr=np.array([[2,7],[7,10]])
eg_val,eg_vect=linalg.eig(arr)
print(eg_val)
print(eg_vect)
```

Output:

```
[-2.06225775+0.j 14.06225775+0.j]
[[-0.86491009 -0.50192682]
[ 0.50192682 -0.86491009]]
```

Integration using scipy library:

PROGRAM:

```
from scipy import integrate
from math import sqrt
a=lambda x:x**4
b=integrate.quad(a,1,6)
print(b)
```

Output:

```
(1555.0, 1.7263968032921184e-11)
```